

$$\int_1^2 \frac{e^x(x-1)}{(x+1)^3}dx$$

$$u:=x+1$$

$$du=dx$$

$$\int_2^3 \frac{e^{u-1}(u-2)}{u^3}du$$

$$\int_2^3 \frac{e^{u-1}u}{u^3}du-2\int_2^3 \frac{e^{u-1}}{u^3}du$$

$$\int_2^3 \frac{e^{u-1}}{u^2}du-2\int_2^3 \frac{e^{u-1}}{u^3}du$$

$$w:=\frac{1}{u^2} \; dv:=e^{u-1}du$$

$$dw=-\frac{2}{u^3}du \; v:=e^{u-1}$$

$$\frac{e^{u-1}}{u^2} \Big|_2^3+2\int_2^3 \frac{e^{u-1}}{u^3}du-2\int_1^2 \frac{e^{u-1}}{u^3}du$$

$$\frac{e^2}{9}-\frac{e}{4}$$